

ADHDNet: A Lightweight Adaptive Multimodal 3D CNN for ADHD Classification Using Structural MRI and Phenotypic Features

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Abstract. ADHD is diagnosed primarily through behavioral and clinical assessment, motivating neuroimaging-based computational approaches. Deep learning models for structural MRI, however, often rely on computationally intensive volumetric architectures, and multimodal approaches add further fusion complexity. This raises the question of whether comparable predictive performance can be obtained from a compact multimodal 3D architecture at substantially lower computational cost. To investigate it, we propose ADHDNet, a lightweight multimodal 3D CNN combining depthwise-separable convolutions and efficient channel recalibration with sample-adaptive fusion of MRI and phenotypic representations. On 497 ADHD-200 subjects from five sites, ADHDNet (71,552 parameters; 0.19 GFLOPs per volume) achieved 90.00% accuracy and an AUC of 0.9543 on a held-out validation set and $87.52 \pm 4.61\%$ accuracy (AUC 0.951 ± 0.017) in 5-fold cross-validation. Relative to a conventional Conv3D multimodal model, it required 94.8% fewer parameters and 97.2% fewer FLOPs at comparable cross-validated AUC. A controlled ablation, however, showed that the MRI-only variant performed close to chance. The results show that comparable multimodal predictive performance can be obtained at substantially lower computational cost, but the ablation also reveals that performance is dominated by phenotypic information, leaving the incremental contribution of structural MRI unresolved.

Keywords: ADHD · Structural MRI · Multimodal learning · 3D CNN · Lightweight deep learning

1 Introduction

Attention-Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder characterized by persistent inattention, hyperactivity, and impulsivity. Its diagnosis relies primarily on behavioral assessment and clinical evaluation, which is challenging because ADHD symptoms overlap with those of other developmental and psychiatric conditions. These challenges have motivated the investigation of neuroimaging-based computational approaches for identifying brain patterns associated with ADHD [1]–[3].

Structural magnetic resonance imaging (sMRI) provides non-invasive information about brain anatomy and has been widely used for computational ADHD analysis [4]. Earlier studies applied conventional classifiers to measures such as gray-matter volume and cortical thickness [5]. Deep learning has since enabled imaging representations to be learned directly from data, with recent ADHD studies exploring convolutional, multimodal, attention-based, and ensemble architectures [6]–[9]. Multimodal approaches further combine imaging with phenotypic information such as age, sex, and clinical ratings. However, increasing depth, channel dimensionality, and fusion complexity can substantially raise the computational and memory requirements of volumetric models, which matters in resource-constrained settings where compact models, such as the few-thousand-parameter network of Deepika et al. [10], have been explored.

It therefore remains unclear whether volumetric structural MRI and phenotypic information can be modeled jointly at substantially lower computational cost, whether a compact 3D representation can retain the predictive performance of a conventional volumetric architecture, and whether the resulting performance derives from the imaging modality, the phenotypic variables, or particular architectural components. Rather than proposing new building blocks, this study combines established efficient components and evaluates their combination under controlled conditions. To address these questions, we propose *ADHDNet*, a lightweight multimodal 3D CNN designed to jointly model structural MRI and phenotypic information under a substantially reduced computational budget. The main contributions of this study are:

1. A **lightweight multimodal 3D architecture** for jointly learning volumetric MRI and phenotypic representations, substantially reducing the parameter and computational requirements of a conventional Conv3D-based multimodal model.
2. A **parameter-efficient MRI representation pipeline** combining depthwise-separable 3D convolutions with lightweight channel recalibration, designed to preserve volumetric feature extraction while limiting model complexity.
3. A **sample-adaptive multimodal fusion mechanism** together with a controlled comparison against concatenation-based fusion, enabling the contribution of adaptive modality integration to be assessed directly.
4. A **systematic evaluation** of predictive performance, model complexity, computational cost, and component contributions using held-out evaluation, 5-fold cross-validation, and controlled ablation experiments.

2 Related Work

2.1 MRI-Based ADHD Classification

Early MRI-based ADHD studies applied conventional classifiers to handcrafted features [1]–[3]; Lohani and Rana [5] combined gray-matter volume and cortical-thickness features with personal characteristics and achieved 75% accuracy with

repeated ten-fold crossvalidation. More recent work learns representations directly: CNNs have been applied to fMRI [6], [11], graph-based models of resting-state networks have been combined with ensemble classifiers or attention [12], [13], and Pedrollo et al. [9] reported CNN ensemble-based classification of resting-state fMRI. These studies demonstrate the feasibility of learning ADHD-related representations, but their emphasis is predictive performance; parameter count, FLOPs, memory footprint, and latency are less consistently reported, making it difficult to assess whether volumetric models can operate under constrained computational budgets.

2.2 Multimodal ADHD Classification

Phenotypic variables such as age, sex, medication status, and clinical ratings can carry information that is not directly represented in image features. Deepika et al. [14] combined resting-state functional connectivity with phenotypic data and reported that adding phenotypic information raised accuracy to 77.59%. A recent preprint has also combined 3D structural MRI with clinical phenotypes, reporting 87% accuracy on the Peking-1 subset of ADHD-200. Although these studies indicate that phenotypic information can complement neuroimaging features, they generally do not isolate the contribution of each modality under a common architecture. Consequently, improvements attributed to multimodal learning may reflect the predictive strength of phenotypic variables rather than complementary information from the imaging modality.

2.3 Lightweight and Parameter-Efficient Models

Depthwise-separable convolution [15], [16] factorizes spatial filtering and channel mixing and is a standard means of reducing the parameter count of convolutional networks, while efficient channel attention (ECA) [17] provides channel recalibration with a negligible number of parameters. In ADHD classification, compact models have so far operated on derived features rather than volumetric images [10]. The combination of these mechanisms within a compact volumetric multimodal architecture for ADHD, with explicit measurement of the resulting computational trade-offs, has not been systematically evaluated.

2.4 Adaptive Multimodal Fusion

Whereas concatenation combines modality features identically for every sample, gated multimodal units [18] learn multiplicative gates that weight each modality per sample. Whether such adaptive fusion provides an advantage over concatenation for MRI and phenotypic data in ADHD has, to our knowledge, not been examined under controlled conditions; this study investigates that question rather than presupposing it.

Table 1. Selected recent ADHD classification studies on ADHD-200 data. Accuracies are not directly comparable because modalities and protocols differ.

Study	Year	Modality	Model	Params	Acc. (%)	Limitations
Lohani & Rana [5]	2023	sMRI + personal char.	SVM/RF/KNN	NR	75.0	Handcrafted features
Gülhan & Özmen [11]	2024	fMRI	3D CNN	NR	76.19 [†]	fMRI only; site-dependent accuracy
Deepika et al. [14]	2024	rs-fMRI + phenotypic	Multimodality model	NR	77.59	Atlas-dependent; no efficiency analysis
Zamanzadeh et al. [12]	2024	rs-fMRI	EasyEnsemble	NR	74.30	Handcrafted graph features
Wu et al. [13]	2025	rs-fMRI	HAGCN	NR	77.95	Requires network construction
Deepika et al. [10]	2025	MRI-derived features	Lightweight attention-KAN	Few thousand	79.25	Derived features, not volumes
ADHDNet (ours)	2026	3D sMRI + phenotype	DS-Conv3D + ECA + gated fusion	0.0716M	90.0[‡]	Phenotype-driven; no external test

NR: not reported. [†]Best per-site accuracy (NeuroIMAGE 76.19%, NYU 69.92%, PU 70.77%).[‡]Held-out split; 5-fold cross-validation: $87.52 \pm 4.61\%$.

2.5 Research Gap

Existing ADHD studies (Table 1) demonstrate the utility of neuroimaging, phenotypic information, and complex deep architectures, but their computational requirements and the individual contributions of modalities and architectural components are not consistently characterized. Three questions remain insufficiently examined: (i) whether structural MRI and phenotypic information can be jointly modeled using a substantially more compact volumetric architecture; (ii) whether parameter-efficient 3D representations can retain predictive performance comparable to conventional volumetric models; and (iii) which architectural components and modalities contribute to the observed performance. ADHDNet is therefore designed not only to provide a lightweight multimodal architecture (questions i–ii; contributions 1–2), but also to experimentally isolate these factors through controlled comparisons and ablation (question iii; contributions 3–4).

3 Materials and Methods

3.1 Dataset

Experiments used the preprocessed anatomical component of the publicly available ADHD-200 dataset [19], [20], comprising 575 successfully processed subjects (222 ADHD, 353 controls) from six sites. The Pittsburgh site contained 78 controls but no ADHD subjects; since it does not provide both classes required for binary discrimination, it was excluded. After excluding Pittsburgh, the final

Table 2. ADHD-200 anatomical data by site; Pittsburgh was excluded (no ADHD subjects).

	KKI	NYU	OHSU	Peking_1	NeuroIMAGE	Pittsburgh	Final
ADHD	22	115	36	24	25	0	222
Control	60	96	35	61	23	78	275
Total	82	211	71	85	48	78	497
ADHD (%)	26.8	54.5	50.7	28.2	52.1	0.0	44.7
Status	Incl.	Incl.	Incl.	Incl.	Incl.	Excl.	

analytical cohort comprised 497 subjects (222 ADHD, 275 controls) across five sites (Table 2).

3.2 Preprocessing and Phenotypic Features

Each NIfTI volume was resized to $64 \times 64 \times 64$ voxels by trilinear interpolation and min-max normalized to $[0, 1]$. No additional skull stripping, registration, bias correction, or segmentation was applied, to keep the pipeline compact and reproducible. Five phenotypic variables were used: age, sex, medication status, acquisition site, and ADHD Index. The latter is a clinical symptom measure closely related to the diagnostic label and is therefore treated as a phenotypic predictor rather than an imaging-derived feature. Missing values (coded -999) were imputed with the training-set median (continuous) or mode (categorical); age and ADHD Index were z-scored with training-set statistics, and categorical variables were scaled to $[0, 1]$.

4 Proposed ADHDNet

ADHDNet (Fig. 1; 71,552 parameters) comprises a 3D MRI encoder, a phenotypic encoder, gated fusion, and a classification head.

Lightweight 3D MRI Encoder. Volumetric encoders are the dominant source of parameters and computation in 3D CNNs, because a standard convolution with kernel size K requires $K^3 C_{in} C_{out}$ weights. Let $X \in \mathbb{R}^{1 \times 64 \times 64 \times 64}$ denote the normalized sMRI volume. The encoder transforms X through three blocks that produce progressively lower-resolution representations with increasing channel capacity ($24 \rightarrow 48 \rightarrow 96$ channels; $64^3 \rightarrow 8^3$). Each block factorizes spatial filtering and channel mixing into a $3 \times 3 \times 3$ depthwise and a $1 \times 1 \times 1$ pointwise convolution [15], [16], followed by batch normalization [21], ReLU, and $2 \times 2 \times 2$ max pooling:

$$H_l = \text{MaxPool}(\text{ReLU}(\text{BN}(\text{PW}_l(\text{DW}_l(H_{l-1}))))), \quad H_0 = X, \quad (1)$$

which requires $K^3 C_{in} + C_{in} C_{out}$ weights per block. This design retains three-dimensional spatial structure while substantially reducing parameter growth;

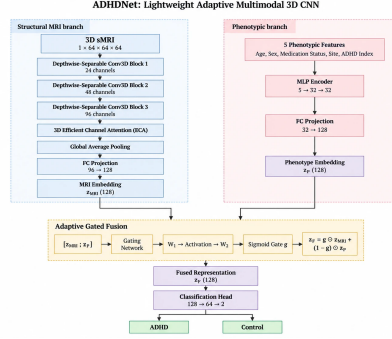


Fig. 1. Architecture of ADHDNet: a depthwise-separable 3D MRI encoder with 3D ECA, an MLP phenotypic encoder, and adaptive gated fusion followed by a classification head.

its contribution to parameter efficiency and predictive performance is evaluated through ablation A4.

3D Efficient Channel Attention. Conventional attention modules add fully connected bottlenecks that are costly in a compact model. To examine whether lightweight channel recalibration can improve the compact volumetric representation without materially increasing model complexity, ADHDNet applies a 3D extension of ECA [17] to the encoder output $F = H_3 \in \mathbb{R}^{96 \times 8^3}$:

$$F_{att} = \sigma(\text{Conv1D}_k(\text{GAP}(F))) \odot F, \quad (2)$$

where GAP denotes global average pooling and the 1D convolution across channels has an adaptively chosen kernel size ($k = 3$ for 96 channels), so that the module adds only three parameters. F_{att} is globally average-pooled and linearly projected to the MRI embedding $z_{MRI} \in \mathbb{R}^{128}$. The contribution of this module is quantified by ablation A1.

Phenotypic Encoder and Adaptive Gated Fusion. Concatenation combines the two modalities identically for every subject, although their informativeness may vary between subjects. To investigate whether modality weighting can be adapted at the subject level, ADHDNet employs a learnable gating function that estimates the relative contribution of the two modalities from their joint representation. The phenotypic branch first maps the five input variables to $h_P \in \mathbb{R}^{32}$ with a $5 \rightarrow 32 \rightarrow 32$ MLP using batch normalization, ReLU, and dropout [22]. Following gated multimodal units [18], a gating network computes a sample-specific vector from the 160-dimensional concatenation, and h_P is projected to $z_P \in \mathbb{R}^{128}$:

$$g = \sigma(W_2 \delta(W_1[z_{MRI}; h_P])) \in \mathbb{R}^{128}, \quad z_F = g \odot z_{MRI} + (1 - g) \odot z_P, \quad (3)$$

where δ is the ReLU. A $128 \rightarrow 64 \rightarrow 2$ head with dropout maps z_F to class probabilities. The fusion strategy is compared with concatenation in ablation A3, and the phenotypic branch as a whole is removed in A2.

5 Experimental Setup

Training. All models were implemented in PyTorch [23] and trained using AdamW [24] (learning rate 10^{-4} , weight decay 10^{-4}), batch size 8, at most 150 epochs with early stopping (patience 15), class-weighted cross-entropy with label smoothing [25] of 0.1, gradient clipping at 1.0, and cosine annealing [26]. Volumes were randomly flipped per axis ($p = 0.5$) during training, the best-validation-AUC checkpoint was retained, and seeds were fixed to 42.

Comparator. A conventional Conv3D multimodal model was implemented within the same framework and trained using the same preprocessing, data splits, fusion mechanism, optimization strategy, and evaluation protocol, so that differences in computational cost and predictive performance can be attributed primarily to its larger volumetric backbone and embedding dimensions. It uses four standard Conv3D blocks ($32 \rightarrow 64 \rightarrow 128 \rightarrow 256$ channels) with 3D ECA after blocks 2–4, generalized-mean pooling, a 256-dimensional embedding, a $5 \rightarrow 32 \rightarrow 64 \rightarrow 64$ phenotypic MLP, the same gated fusion, and a $256 \rightarrow 128 \rightarrow 64 \rightarrow 2$ head (1,376,080 parameters).

Evaluation Protocol. (i) *Held-out validation:* a stratified 80:20 split yielded 397 training and 100 validation subjects (45 ADHD, 55 controls). Because the validation set was also used for early stopping and model selection, these results represent held-out validation rather than independent test performance. (ii) *Cross-validation:* to determine whether the held-out estimate is representative, both models were evaluated with stratified 5-fold cross-validation on all 497 subjects (max. 60 epochs, patience 10; metrics at the best-AUC epoch). (iii) *Ablation:* to isolate the contribution of depthwise-separable convolution, channel attention, the phenotypic branch, and adaptive fusion, each variant modifies one component while retaining the remaining architecture, preprocessing, optimization procedure, and split. Imputation and normalization statistics were always fitted on training subjects only, and ADHD was the positive class. FLOPs were counted with PyTorch’s `FlopCounterMode` for one 64^3 volume, and latency was averaged over 50 runs after 10 warm-up passes on a CUDA GPU.

6 Results

6.1 Held-Out Classification Performance

On the held-out validation set (Table 3), ADHDNet achieved 90.00% accuracy and an AUC of 0.9543; the best validation AUC was reached at epoch 60 and training stopped at epoch 75. It correctly classified 43 of 45 ADHD subjects and

Table 3. Held-out validation performance of ADHDNet ($n = 100$; ADHD positive).

Accuracy	AUC	Sensitivity	Specificity	Precision	NPV	F1-score
90.00%	0.9543	95.56%	85.45%	84.31%	95.92%	0.8958

**Fig. 2.** Held-out validation results ($n = 100$): (a) ROC curve (AUC = 0.9543); (b) confusion matrix at threshold 0.5.

47 of 55 controls (Fig. 2), with 8 false positives and 2 false negatives; sensitivity was 95.56%, specificity 85.45%, precision 84.31%, and NPV 95.92%.

6.2 Cross-Validation and Computational Efficiency

Table 4 compares ADHDNet with the conventional Conv3D comparator under identical data and training conditions. In 5-fold cross-validation, ADHDNet achieved $87.52 \pm 4.61\%$ accuracy (per-fold range 79.80–90.91%) and an AUC of 0.9512 ± 0.0169 , against $86.51 \pm 3.64\%$ and 0.9566 ± 0.0197 for the comparator (Fig. 3a). The models showed comparable cross-validated discrimination, with ADHDNet yielding a slightly higher mean accuracy, sensitivity, and F1-score but a slightly lower mean AUC. The held-out accuracy of 90.00% lies at the upper end of the cross-validated range, so 87.52% is the more representative estimate. ADHDNet required 94.8% fewer parameters ($19.2\times$) and 97.2% fewer FLOPs ($35.2\times$), reducing FP32 weight storage from 5.25 to 0.27 MiB (Fig. 3b). Latency decreased by a factor of 1.8 and batched throughput increased by a factor of 2.7. Substantially reducing the volumetric backbone therefore did not cost multimodal predictive performance in this setting, although, as the ablation shows, this performance is not primarily attributable to the MRI representation.

6.3 Ablation Study

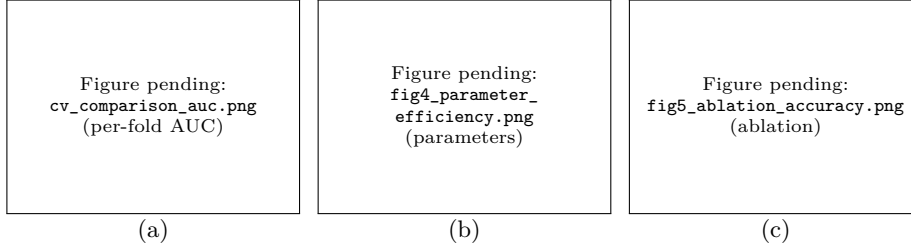
The ablation variants (Table 5, Fig. 3c) remove ECA (A1), remove the phenotypic branch (A2), replace gated fusion with concatenation (A3), or use standard convolutions (A4). Removing the phenotypic branch reduced performance to near chance (AUC 0.5644), whereas the other variants changed accuracy by at most 3 points. With 100 validation subjects, one subject equals one percentage point, so small differences should be interpreted with caution.

Table 4. ADHDNet versus the Conv3D comparator: complexity per 64^3 volume (latency at batch 1, throughput at batch 8) and 5-fold CV (mean $\pm$ SD).

	Measure	Conv3D comparator	ADHDNet
Complexity	Parameters	1,376,080	71,552
	FP32 weights	5.25 MiB	0.27 MiB
	GFLOPs	6.7952	0.1932
	Latency (ms)	4.32 ± 0.12	2.36 ± 0.15
	Throughput (vol/s)	242.6	662.9
5-fold CV	Accuracy (%)	86.51 ± 3.64	87.52 ± 4.61
	AUC	0.9566 ± 0.0197	0.9512 ± 0.0169
	Sensitivity (%)	85.10 ± 9.63	87.36 ± 5.75
	Specificity (%)	87.64 ± 7.09	87.64 ± 4.71
	F1-score (%)	84.79 ± 4.70	86.21 ± 5.20

Table 5. Component-wise ablation on the held-out validation set ($n = 100$).

Variant	ECA	Pheno.	Fusion	Params	Acc. (%)	AUC	F1
A0 Full ADHDNet	✓	✓	Gated	71,552	90.00	0.9543	0.8958
A1 w/o ECA	–	✓	Gated	71,549	87.00	0.9487	0.8632
A2 w/o phenotype	✓	–	–	28,896	54.00	0.5644	0.4773
A3 Concat. fusion	✓	✓	Concat.	66,272	89.00	0.9564	0.8842
A4 Standard Conv3D	✓	✓	Gated	220,133	89.00	0.9547	0.8842

**Fig. 3.** (a) Per-fold CV AUC; (b) parameters of ADHDNet and the comparator; (c) ablation accuracy.

7 Discussion

ADHDNet matched the comparator’s cross-validated discrimination at a fraction of its cost, but this performance is dominated by phenotypic rather than structural MRI information.

Parameter Efficiency. Replacing depthwise-separable with standard convolutions (A4) substantially increased the parameter count without improving accuracy or AUC on the evaluated split, suggesting that the additional convolutional capacity was not beneficial under this experimental setting.

Contribution of Channel Attention. The 3.0-point drop without ECA (A1) corresponds to three validation subjects, so ECA is best regarded as a low-cost refinement whose benefit requires confirmation with repeated runs.

Contribution of Multimodal Fusion. Without the phenotypic branch (A2), performance was close to chance, whereas every variant retaining it reached an AUC above 0.94. Adaptive gating did not consistently improve over concatenation (A3), so its benefit remains to be established.

What the Results Do Not Demonstrate. The phenotypic input includes the ADHD Index, a clinical symptom rating closely related to the diagnostic label, and site, whose class proportions differ markedly (Table 2). The reported accuracy therefore does not demonstrate that structural MRI, as represented here, discriminates ADHD, nor should it be interpreted as clinically applicable diagnostic performance.

Comparison with Prior Work. ADHDNet’s accuracy exceeds those in Table 1 numerically, but the studies differ in modality and protocol. The primary distinction of this study is therefore not a claim of superior classification accuracy, but the combination of compact volumetric multimodal processing with controlled analysis of computational cost and component contributions.

Limitations and Future Work. Performance depends predominantly on phenotypic features; metrics were taken at the best-validation-AUC epoch without an external test set; cross-site generalization was not evaluated; the ablation used a single split without repeated runs; the simple preprocessing may limit the anatomical information available to the MRI branch; and only a single compact architecture was evaluated, so the observed efficiency–performance trade-off is not established for other lightweight volumetric backbones.

8 Conclusion

This study investigated whether structural MRI and phenotypic information can be jointly modeled using a substantially more compact 3D architecture. ADHDNet achieved cross-validated discrimination comparable to a conventional Conv3D multimodal model while using 94.8% fewer parameters and 97.2% fewer FLOPs. The component analysis further showed that depthwise-separable convolution provided the principal reduction in model complexity, whereas channel recalibration produced only a modest gain and adaptive gating did not consistently outperform concatenation. Most importantly, removing the phenotypic branch reduced performance to near-chance levels, indicating that the reported multimodal performance is dominated by phenotypic information in the present setting. Consequently, the incremental contribution of structural MRI remains unresolved. Future evaluation should remove potentially label-proximal variables such as the ADHD Index and site, and assess generalization across sites and external cohorts.

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